GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Statistics for Data Science

DS-501-MJ

Interview Preparation — Topic-wise Q&A

Semester I · Core

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Curated interview questions and answers for Statistics for Data Science, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Descriptive statistics

Q. What is the difference between mean, median and mode?

Mean is the arithmetic average, median is the middle value of ordered data, and mode is the most frequent value. Median is preferred for skewed data because it is robust to outliers.

Q. What do variance and standard deviation measure?

Both measure dispersion around the mean. Variance is the average squared deviation; standard deviation is its square root, expressed in the same units as the data.

Q. What is skewness and kurtosis?

Skewness measures asymmetry of a distribution (positive = long right tail). Kurtosis measures 'tailedness' — high kurtosis means heavy tails and more outliers.

Correlation & regression

Q. What does the correlation coefficient r indicate?

r lies between -1 and +1 and measures the strength and direction of a linear relationship. r=0 means no linear correlation. Correlation does not imply causation.

Q. Difference between correlation and regression?

Correlation quantifies the strength of association between two variables; regression models the relationship to predict one variable from another (Y = a + bX).

Q. What is R-squared?

The proportion of variance in the dependent variable explained by the model. Adjusted R-squared penalises adding non-useful predictors.

Q. What is multicollinearity?

When predictor variables are highly correlated with each other, making coefficient estimates unstable. Detected via VIF; fixed by dropping/combining variables.

Probability & hypothesis testing

Q. State Bayes' theorem.

$P(A|B) = P(B|A) \cdot P(A) / P(B)$. It updates the probability of a hypothesis given new evidence.

Q. What is a p-value?

The probability of observing the data (or more extreme) if the null hypothesis were true. If p < significance level (e.g. 0.05), we reject the null hypothesis.

Q. When do you use a t-test vs a z-test?

Use a z-test for large samples with known variance; a t-test for small samples with unknown variance.

Q. What is ANOVA used for?

To test whether the means of three or more groups differ significantly, using the F-statistic.

Q. Explain the Central Limit Theorem.

The sampling distribution of the sample mean approaches a normal distribution as sample size grows, regardless of the population's distribution.